Anelench, explained

A tool that asks whether a feature in a black-hole image is really demanded by the telescope data, and plans the observation that would settle it

Built for Banana Hacks 2026 (Image AI). This document explains the project in plain words; the companion document "Anelench: mathematics, algorithms and physics" gives every formula and proof.

1. The short version

The famous image of the black hole M87* is not a photograph. A handful of radio telescopes spread over the Earth measured a few hundred numbers, and an algorithm filled in everything between them. Change the algorithm's assumptions and the fine details of the image change. So when a researcher sees a faint bright spot next to the ring, there is a fair question to ask: is the spot really in the sky, or did the algorithm put it there?

Anelench answers that question with images, in four steps:

1. **Mark.** You draw a circle around the structure you doubt and state a claim, for example "this region is at least 20% as bright as the ring's brightest part".
2. **Test.** Anelench looks for two images that both explain the measurements equally well: one in which the claim is true and one in which it is false. It searches under seven different imaging assumptions, including two generative models we trained on simulated black holes.
3. **Plan.** If both images exist, the data cannot decide. Anelench then scores every observing night the real telescopes could actually carry out (which telescopes, which hours, how sensitive, how expensive) by how strongly that night would tell the two images apart, and recommends the best one per hour of telescope time.
4. **Check.** It adds the recommended night (simulated from a hidden true image in tests, or a real night that we held back) and runs the whole test again, so its own recommendation is verified, not trusted.

2. Why this is a real problem

2.1 How the Event Horizon Telescope makes an image

No single telescope is large enough to see the shadow of M87*: it is about 40 microarcseconds (μas) across, the size of a doughnut on the Moon seen from Earth. The Event Horizon Telescope (EHT) combines radio dishes in Chile, Mexico, Arizona, Hawai'i, Spain and elsewhere into one Earth-sized virtual telescope. Each pair of dishes measures one "visibility": one sample of the two-dimensional Fourier transform of the sky, a number that says how much structure the sky has at one particular scale and orientation. As the Earth turns, each pair sweeps a short track of such samples.

In April 2017 the array collected only a few hundred useful samples per night. An image needs thousands of pixels. The gap is filled by an imaging algorithm, and every imaging algorithm has to assume something about what images usually look like: smooth, sparse, ring-like, similar to a training set. Where the data are silent, the assumption speaks.

2.2 Why it matters now

Recent work shows how much these choices matter. Feng, Bouman and Freeman (2024) showed that different learned priors change the features and the uncertainties of the M87* reconstruction. Dahale and colleagues (2025) found that the ring's ellipticity in simulations does not track the black hole's spin but the structure of the emitting gas: measuring a shape in an image and explaining its physical cause are separate steps. Anelench puts a measurable check between the two: before a feature is interpreted, test whether the data demand it.

3. What Anelench does, step by step

3.1 Mark a feature and make a claim

You pick a dataset and an imaging assumption, see the reconstruction, and click on the structure you want to test. The claim is always a number compared with a threshold: the mean brightness inside your circle divided by the peak brightness of the image, both after blurring the image to a common 20 μas resolution. The blurring matters: a sharp image and a smooth image must not "differ" only because one is sharper. Dividing by the peak matters too: the measurements cannot fix the total brightness, so a fair feature measure must not depend on it.

3.2 Look for two explanations

Anelench now tries to break your claim and to strengthen it. For each of the seven imaging assumptions it takes the reconstruction and pushes it twice: once to make the region as faint as possible, once to make it as bright as possible, while the image must keep fitting the measurements within the noise. If it finds an acceptable image on each side of your threshold, the claim is **undetermined**: the same data are explained equally well with and without the feature. You see the two images side by side and in a blink comparator (the instrument Clyde Tombaugh used to find Pluto: the view flips between the two images so the difference jumps out), and you see both images' predictions pass through the same measured data points.

If only one side can be reached, the verdict is **supported** or **not supported**. These are weaker statements than "undetermined": they mean no counterexample was found under the assumptions tried, not that none exists. An "undetermined" verdict, in contrast, is proven by two explicit images.

3.3 Plan the next observation

If the claim is undetermined, more data are needed. But which data? Anelench builds 173 feasible follow-up nights from real stations: the 2017 array, with and without the telescopes that joined later (the Greenland Telescope, Kitt Peak and NOEMA), in three- and six-hour windows at different times of day. For each night it computes, from the Earth's rotation and the stations' positions, which pairs of telescopes see M87* above 10° elevation and which Fourier samples they would measure, and how noisy those samples would be given each telescope's sensitivity. It then predicts what each pair of competing images would make that night measure. A good night is one where the images predict clearly different values, so that whichever is wrong would stand out. Anelench asks every night to separate even the hardest pair, and divides that separation by the night's cost in telescope-hours.

3.4 Check the recommendation

The ranking is a prediction made with a fixed set of images. Once new data arrive, the images can adapt. So Anelench does not stop at a recommendation: it adds the night, re-runs every imaging assumption and the whole search, and reports how the range of acceptable answers changed. On synthetic tests the night is simulated from the hidden true image, with fresh telescope errors and noise; on the real M87* data, it is one of the 2017 nights we held back.

4. The ideas inside, in plain words

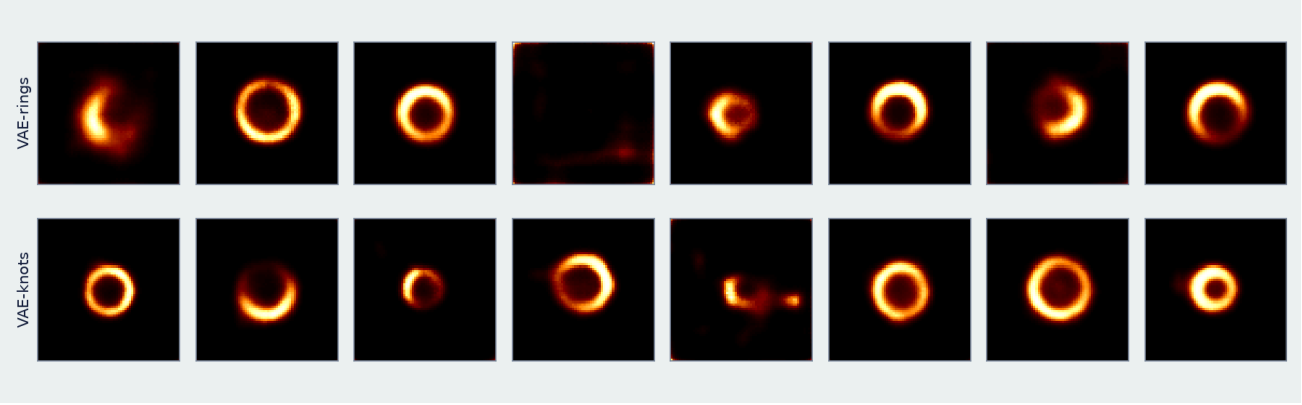
4.1 Measurements that survive bad calibration

Each telescope adds its own error to everything it measures: at 1.3 mm wavelength the atmosphere above a dish shifts the signal's phase unpredictably within seconds. Anelench therefore never uses single measurements directly. It combines them so that every telescope's error cancels. Multiply the three measurements around a triangle of telescopes and each telescope's phase error appears once with a plus and once with a minus sign: the "closure phase" of the triangle is error-free. A similar ratio of four amplitudes, the "closure amplitude", cancels each telescope's amplitude error. Anelench uses only these quantities, chooses a set in which none is a combination of the others, and prefers the least noisy ones.

4.2 Seven ways to draw an image

Each "imaging assumption" is a different machine that turns a list of numbers into an image, together with a preference for some images over others:

Assumption	What it is	What it prefers
RML-MEM	free pixels, the classic EHT method	images close to a smooth blob, without roughness
RML-TV	free pixels	flat patches with sharp edges
NF-1.5	a neural field: a small neural network that maps a position on the sky to a brightness	smooth structure
NF-3.0	the same with higher-frequency inputs	compact, finer structure
DIP	a "deep image prior": an untrained convolutional network whose architecture is the preference	natural-looking images
VAE-rings	a generative model we trained on 24,000 simulated black-hole rings, none with a knot	rings; it has never seen a knot
VAE-knots	the same model trained on images half of which have a knot	rings, sometimes with a knot



Random draws from the two generative models we trained. Top: the model that has only seen rings. Bottom: the model whose training set contained knots.

The two trained models are the "learned priors" of the project. Searching only among images such a model can draw turned out to be too rigid to fit the real measurements, so Anelench lets the image change freely but makes it pay for every difference from the closest image the model can draw. With this, the learned priors give the cleanest rings of all seven assumptions, and the ring-only model visibly resists drawing a knot. That is precisely why no single assumption is allowed to decide a claim alone.

4.3 What "fits the data equally well" means

Every measurement has noise, so no image fits perfectly. The usual measure of misfit is χ^2: the sum of squared differences between predicted and measured values, each divided by its noise. A flexible image model can bend itself to fit part of the noise too, so its best fit is better than the true image would be. Anelench measures how much each model does this, by a simple experiment: it treats the model's own image as if it were the truth, adds fresh noise, refits, and sees how far the refit falls below. This "optimism" (26–31 units for our models) matched the true image's actual distance above the best fit (24–34 units) on synthetic data. An image counts as fitting equally well if its misfit is within the optimism plus two of its standard deviations of the best fit any model reached. With this rule the true image is accepted about 97% of the time, and the allowance does not grow just because more data were collected.

4.4 How the search works

Each model is optimised with a goal that rewards moving the region's brightness in one direction, and two "walls" it may not cross: the data-fit allowance above, and a limit on how irregular the image may become by the model's own standard. Several kinds of starting points are used, because an optimiser that starts in one place can miss solutions elsewhere: the model's own reconstructions (each one pushed both ways, which gives "twin" pairs that differ mainly in your region), for pixel models the reconstruction with the region erased or brightened, and after new data the previous counterexamples themselves, which must now also fit the new night.

4.5 How nights are compared

For a candidate night and a pair of competing images, Anelench computes the expected separation: the misfit the wrong image would collect on that night's data, in units of the noise. A separation of 3 means the wrong image would typically be off by three noise levels; such a night favours the right image clearly about 80% of the time, 5 makes it about 98%. Anelench takes the smallest separation over all pairs (a

night must separate every pair), divides by the cost, and recommends the best night within the budget. It shows what simpler strategies would have chosen: the night with the most signal, the night that fills the most empty Fourier space, the night where one model's own uncertainty is largest, and a random night.

5. The data

5.1 Real EHT data

We use the public calibrated data of M87* from the 2017 EHT campaign (release 2019-D01-01): four nights (April 5, 6, 10 and 11), two frequency bands each, from ALMA, APEX, the SMT, JCMT, the LMT, the IRAM 30 m and the SMA. Anelench reads the files with its own reader and checks itself in two ways. First, it recomputes the Fourier coordinates of every measurement from the telescopes' positions and the Earth's rotation; they agree with the files to three parts in a hundred thousand. Second, the bright side of every reconstruction of April 11 is in the south, as in the published EHT images; this caught a sign convention of the files, which our first reconstruction showed upside down. It also recovers each telescope's effective sensitivity from the error bars in the files, which it then uses to predict the noise of future nights.

5.2 Synthetic tests

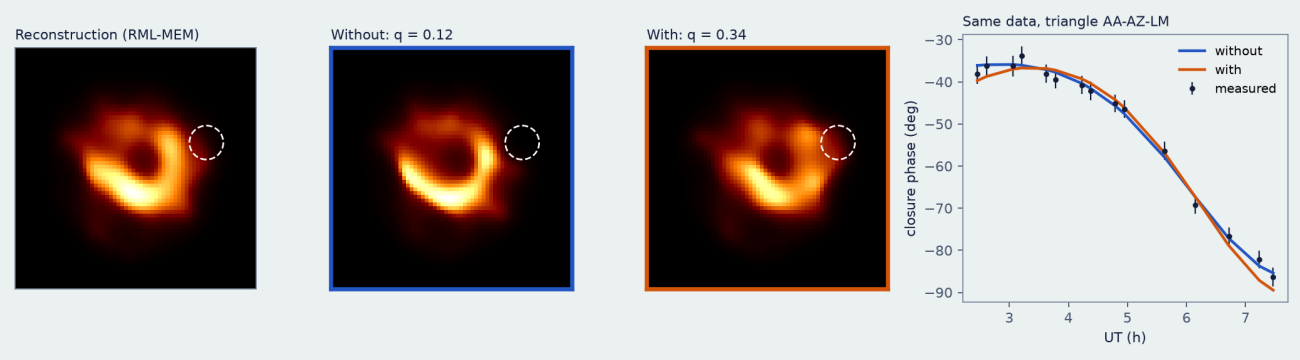
To know whether a verdict is right, we need cases where the truth is known. The synthetic tests are M87*-like rings measured at exactly the same moments and by exactly the same telescope pairs as the real April 11 data, with random telescope errors, a 2% extra error and thermal noise of the right size. Test A has no knot; test B has a faint knot about 40 μas from the centre along the direction of M87's jet. The interface keeps the truth hidden until the end.

6. What we found

6.1 Synthetic test A: a knot that is not there

The true image has no knot (region value 0.08 against a claim threshold of 0.20). Yet five of the seven assumptions draw one: their reconstructions put the region between 0.23 and 0.37. A researcher using any of them would see a knot. Anelench marks the claim **undetermined**: acceptable images span 0.04 to 0.48, and it shows a twin pair from the classic method, with the region at 0.12 and 0.34, whose total misfits differ by only 22 units over 392 measurements.

Anelench recommends a three-hour, ALMA-led night (cost 27 weighted telescope-hours, expected separation 8.5 noise units for the hardest pair); maximum-signal, maximum-coverage and single-model strategies would each spend 69. After the night is added, the range narrows to 0.04–0.41, and the classic reconstruction itself moves to the correct side of the threshold (0.18). A second planned night narrows it to 0.04–0.40. The upper end is held up by two assumptions (the high-bandwidth neural field and the deep image prior) that can still fit a knot within the noise: under those assumptions, closure data alone do not exclude it.

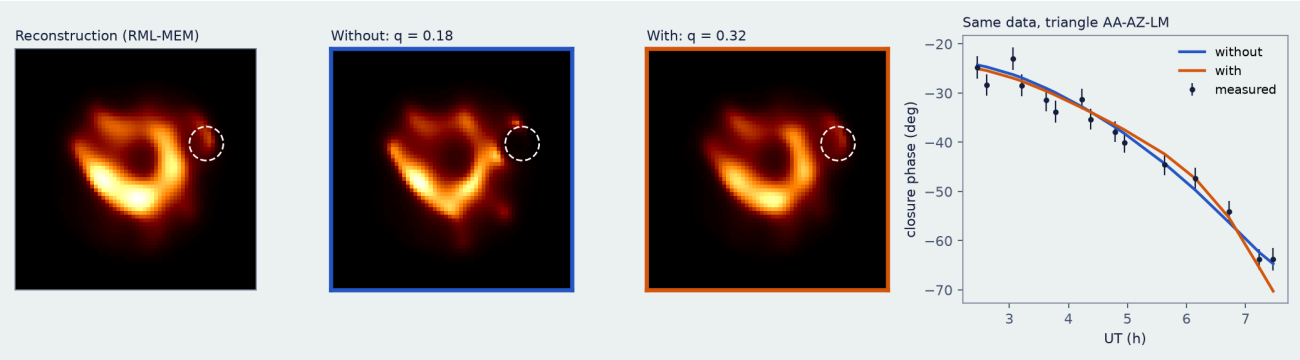


Synthetic test A. Left: the classic reconstruction, which shows a knot in the circle. Middle: two images from the same starting point, without and with the knot. Right: both pass through the same measured closure phases.

6.2 Synthetic test B: a faint knot that is there

Here the knot is real: it carries 6% of the flux and its region value is 0.25, so the claim (above 0.20) is true. The five hand-built assumptions all see it (0.27 to 0.38). The two trained generative models do not: the model that never saw a knot puts the region at 0.14, the one trained with knots at 0.16. This is the effect Feng, Bouman and Freeman warned about, reproduced on purpose: a strong learned prior can erase a real feature. Anelench reports **undetermined** (acceptable images span 0.14 to 0.44), with a twin pair from the classic method at 0.18 and 0.32 whose fits are indistinguishable (their total misfits differ by 0.2 units).

The recommended night adds the Greenland Telescope and Kitt Peak to the 2017 array for three hours (cost 34.5; the coverage strategy would spend 69). With it, the generative models move towards the truth (0.19 and 0.18) and the range narrows to 0.18–0.38; a second planned night gives 0.17–0.39. The low end is held by the knot-free learned prior, which the closure data of two extra nights cannot yet overrule. That is the honest answer: whether this knot is "seen" still depends on whether one admits that prior, and Anelench makes the dependence visible instead of hiding it in a single image.



Synthetic test B. The classic reconstruction and a twin pair without and with the knot; both fit the measured closure phases equally well.

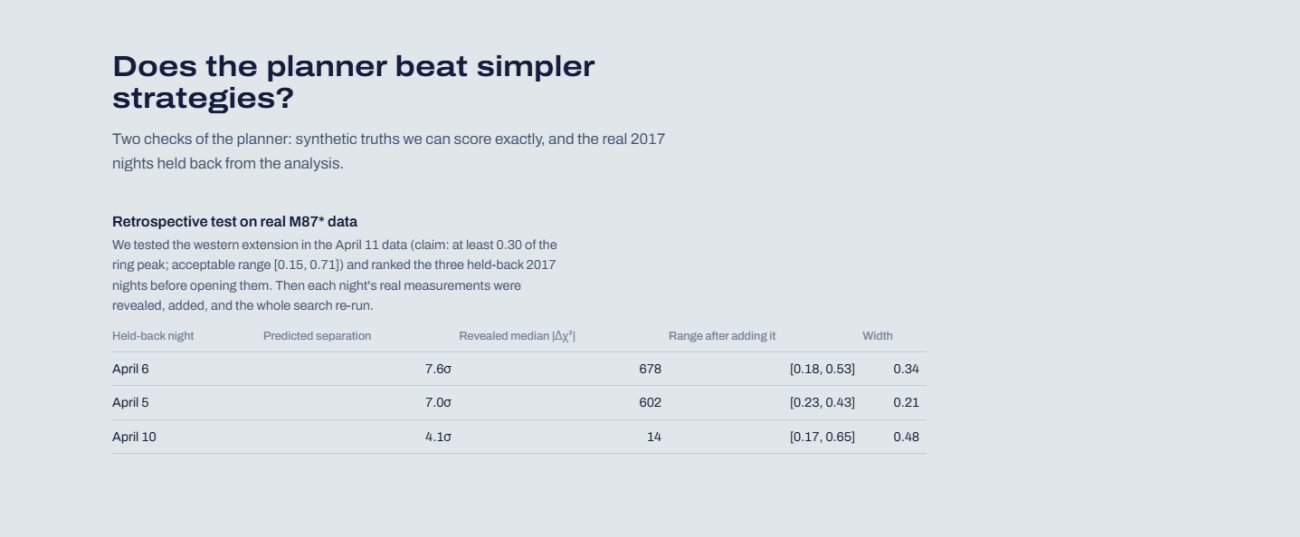
6.3 Real M87*: the western extension

In the real April 11 data, every one of the seven assumptions shows emission about 37 μas west of the centre, near where the jet starts: the region sits at 0.30 to 0.50 of the ring peak in all seven reconstructions, so a claim that it is at least 0.30 would look safe with any single pipeline. Anelench finds acceptable images from 0.16 to 0.71 for the same data: **undetermined**. Its twin pair from the classic

method puts the region at 0.24 and 0.50, and the version with the bright extension even fits slightly better.

To check the planner on real measurements, we held back the other three 2017 nights and ranked them by the predicted separation of the twin pairs before opening them: April 6 (7.6 noise units), April 5 (7.0) and April 10 (4.1). Revealed, the real data separated the explanations in exactly that order (median $\Delta\chi^2$ of 678, 602 and 14), and all three nights favoured the explanation without the bright extension. Adding each night and re-running the whole search narrowed the range most with April 5 (to 0.23–0.43), then April 6 (0.18–0.53), and least with April 10 (0.17–0.65).

Two honest caveats. The per-cost criterion picked April 10, the cheapest night that just clears the 3σ floor, and it was the least informative: dividing by cost rewards nights that barely qualify. And the separations on April 5 and 6 are far larger than predicted, which we read as the source itself changing over six days, as the EHT reported between these nights; those nights test the static-source assumption as much as the extension.



Does the planner beat simpler strategies?

Two checks of the planner: synthetic truths we can score exactly, and the real 2017 nights held back from the analysis.

Retrospective test on real M87* data

We tested the western extension in the April 11 data (claim: at least 0.30 of the ring peak; acceptable range [0.15, 0.71]) and ranked the three held-back 2017 nights before opening them. Then each night's real measurements were revealed, added, and the whole search re-run.

Held-back night	Predicted separation	Revealed median $\lvert\Delta\chi^2\rvert$	Range after adding it	Width
April 6	7.6σ	678	[0.18, 0.53]	0.34
April 5	7.0σ	602	[0.23, 0.43]	0.21
April 10	4.1σ	14	[0.17, 0.65]	0.48

The retrospective test on real data, as shown in the interface: predicted separation, the separation the real data then showed, and the range of acceptable answers after adding each night.

6.4 Across many synthetic truths

Two scenarios can be lucky or unlucky, so we also drew eight random M87*-like truths (four with a faint knot near the jet base, four without), measured each with the real April 2017 sampling and ran the whole procedure, with four of the assumptions to keep the run within a few hours.

On the initial data, a single reconstruction was wrong half the time. The classic method reported a knot in all eight cases, so it was wrong for every knot-free truth. Anelench answered "undetermined" in all eight cases: it confirmed no false claim.

One more night rarely settles the question, whatever the strategy. For each case, each strategy scheduled one night within a budget of 70 weighted telescope-hours, the night was simulated from the truth, and the full search was repeated.

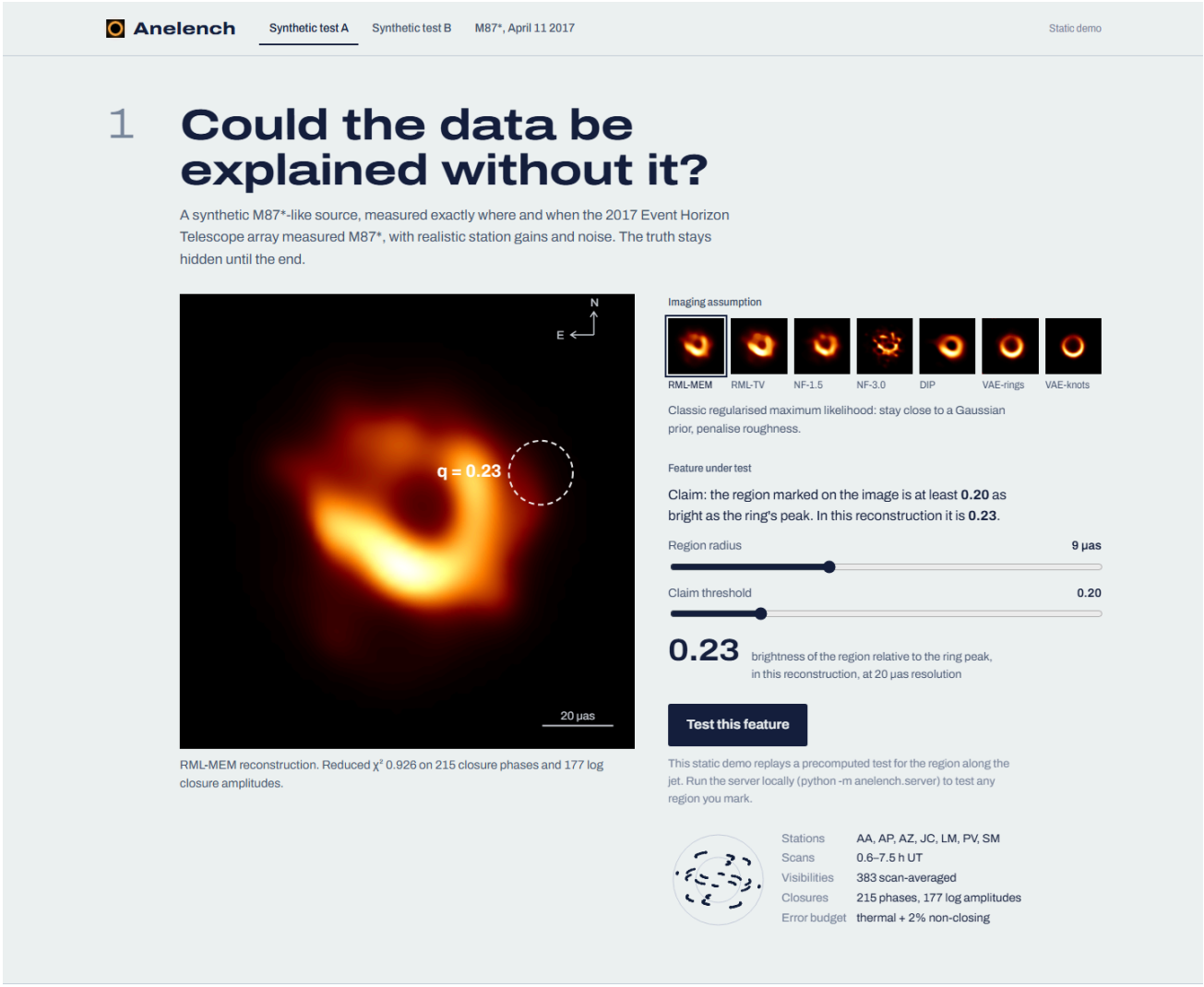
Strategy	Mean cost	Settled correctly	Settled wrongly	Doubt left
Anelench, separation per cost	29	1 of 8	0	0.23
Anelench, whole budget	51	1 of 8	1 of 8	0.19
Highest signal-to-noise	69	1 of 8	0	0.19
Largest new Fourier coverage	69	1 of 8	0	0.17
Single-model uncertainty	55	1 of 8	0	0.20
Random feasible night	38	0 of 8	0	0.28

"Doubt left" is the mean width of the range of acceptable answers after the night. Anelench's per-cost nights reached the same number of correct settlements as the three expensive strategies at 42% of their telescope time, but left somewhat more doubt; spending the whole budget brought its doubt down to their level. Random nights settled nothing and left the most doubt.

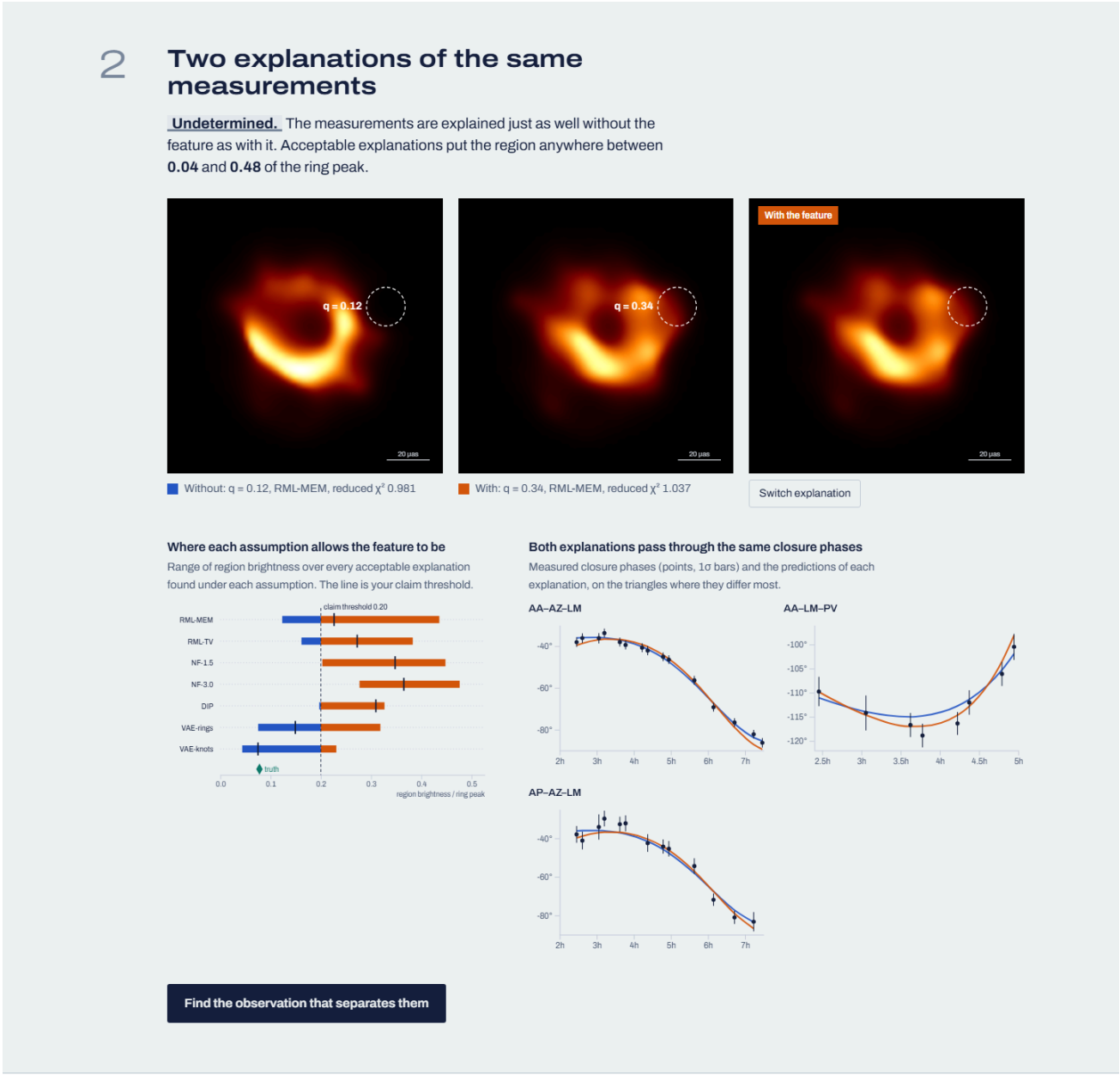
What went wrong, and what it means. In one of the 48 runs a verdict after the new night was wrong: the whole-budget variant called a knot "supported" where there was none, because the search did not reach the knot-free images that the true image proves exist. Relatedly, the reported range contained the truth in only about half of the cases. Both are the same limitation: the search is local, so its range is an inner estimate. "Undetermined" is always backed by two explicit images; "supported" and "not supported" should be read as "no counterexample found".

7. Using the interface

- **Step 1.** Choose a dataset at the top. The large image is the reconstruction under the selected assumption; the thumbnails switch between assumptions. Click the image to move the circle, set its radius and the claim threshold; the number shows the region's value in this reconstruction. Press "Test this feature".
- **Step 2.** The verdict, the two explanations (blue frame: without the feature, orange: with it), the blink comparator, a chart of the range each assumption allows, and the measured closure phases with both explanations' predictions.
- **Step 3.** The globe shows the Earth as seen from M87*: only telescopes on the visible face can observe it; the slider moves through the night. The recommendation, what other strategies would schedule, every feasible night by cost and separation, and the closure phases the recommended night would measure under each explanation.
- **Step 4.** The range of acceptable answers before and after the new night, an option to plan one more night, and, for the synthetic tests, the hidden truth. For M87*, the held-back real nights with their predicted and revealed separations.



Step 1: the reconstruction under the selected assumption, the seven assumptions as thumbnails, the circle and the claim.



Step 2: the verdict, the twin pair, the range each assumption allows (the green diamond marks the truth, shown only after the test), and the closure-phase fits.

3 The observation that tells them apart

Anelench predicted what each of 8 twin witness pairs (each pair grown from one reconstruction, so they differ mainly in the region) would make the array measure on 173 feasible follow-up nights, and ranked the nights by the separation of the hardest pair per unit of telescope time.



Earth as seen from M87*: only stations on the visible face can observe it.

UT 1.5 h

ALMA-led southern, 00:00-03:00 UT

Stations: AA, AP, LM, AZ, PV, NN. 3 hours, cost 27.0 weighted station-hours of a 70 budget.

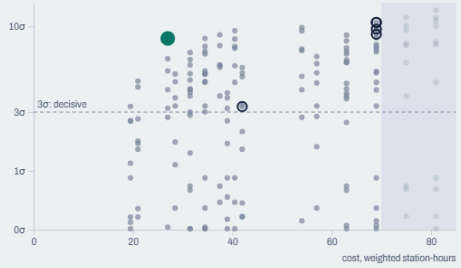
8.5σ expected separation for the hardest pair, across 238 closure phases and 208 log closure amplitudes.

What other strategies would schedule

Strategy	Chosen night	Cost	Hardest pair	Per station-hour
Anelench: separate the hardest witness pair	**ALMA-led southern, 00:00-03:00 UT**	**27.0**	**8.5σ**	**0.32σ**
Anelench, whole budget: largest hardest-pair separation	**2017 + NOEMA, 00:00-06:00 UT**	**69.0**	**10.5σ**	**0.15σ**
Highest total signal-to-noise	2017 + GLT + KP, 02:00-08:00 UT	69.0	9.0σ	0.13σ
Largest new Fourier coverage	2017 + GLT + KP, 04:00-10:00 UT	69.0	9.7σ	0.14σ
Largest single-model uncertainty	2017 + NOEMA, 00:00-06:00 UT	69.0	10.5σ	0.15σ
Random feasible package	Europe-Americas, 04:00-10:00 UT	42.0	3.3σ	0.08σ

Every feasible night, by cost and separation

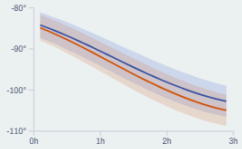
Shaded: over the budget of 70 weighted station-hours. Filled circle: Anelench's choice; rings: the other strategies.



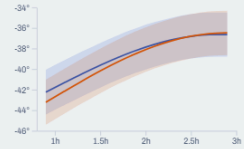
What the recommended night would measure

Predicted closure phases under each explanation, with the expected 1σ noise of the new measurements.

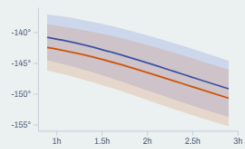
AA–LM–NN, up to 0.6σ apart



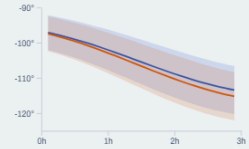
AA–AZ–LM, up to 0.5σ apart



AA–AZ–NN, up to 0.4σ apart



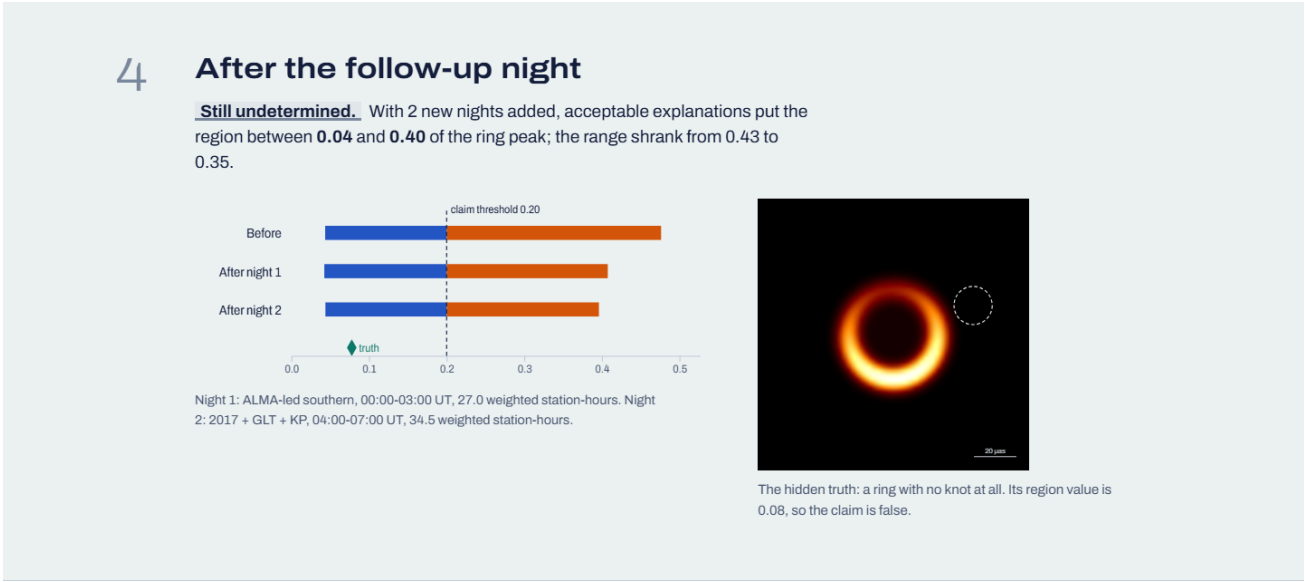
AA–LM–PV, up to 0.3σ apart



Simulate the follow-up night and test again

The follow-up is simulated from the hidden truth with fresh gains and noise; then every assumption is re-imaged and the search runs again.

Step 3: Earth as seen from M87*, the recommended night, the strategy comparison, every feasible night by cost and separation, and the predicted signals.



Step 4: the range of acceptable answers before and after each planned night, and the hidden truth.

8. Running it

On Windows, double-click **run.bat**. On macOS or Linux, run **bash run.sh** in a terminal. The script finds Python 3.10 or newer, installs the required packages into a private folder (.venv) if they are missing, downloads the public EHT data once (about 4.5 MB), starts the engine and opens the page at http://127.0.0.1:8000. The first time a dataset is opened, its seven reconstructions are computed (one to four minutes, shown live); after that they are cached. On a laptop CPU a feature test takes two to four minutes, scoring the nights under a minute, and a re-test about five minutes. Stop the engine with Ctrl+C in its window.

The folder web/ also works on its own as a static website that replays results computed in advance with the same code; it opens instantly and is what the demo video shows.

9. What is in the folder

Path	Purpose
run.bat, run.sh	one-step start on Windows and on macOS or Linux
anelench/data.py	reading the EHT files, averaging, error budget, telescope sensitivities
anelench/closures.py	error-free closure quantities and the choice of an independent set
anelench/geometry.py	Earth rotation, elevations, Fourier coordinates
anelench/forward.py	from an image to predicted closure quantities, and the misfit
anelench/families.py, learned.py	the seven imaging assumptions and the training of the generative models
anelench/feature.py	turning a claim into a number at a common resolution
anelench/search.py	reference images, optimism, the counterexample search
anelench/design.py	feasible nights, noise, scoring, the comparison strategies
anelench/pipeline.py, server.py, serialize.py	putting the steps together, the live engine and the data sent to the page
web/	the interface
models/	the two trained generative models
experiments/, scripts/, tests/	the benchmark, data download, demo builder, figures, model training and the tests of the physics
docs/	this document, the mathematics document and the figures

10. Honest limits

- The search is a local optimisation. The range it reports is an inner estimate: anything it shows is real, but more extreme acceptable images may exist. "Undetermined" is proven; "supported" and "not supported" mean "no counterexample found".
- Only closure quantities are used. They are immune to telescope errors but discard information that careful amplitude calibration could add, so some features stay undetermined for longer than they would in a full EHT analysis.
- The night ranking is computed with a fixed set of competing images. It is a prediction, which is why it is always checked by re-testing.
- Different nights of 2017 are treated as views of the same source. M87* can change over days, so the real-data check also tests that assumption.
- Total intensity of one source only; polarisation and fast variability (as in Sgr A*) are not modelled.

11. Glossary

μas	microarcsecond, one millionth of an arcsecond; M87*'s ring is about 42 μas across.
Visibility	what one pair of telescopes measures: one Fourier component of the sky.
Closure phase / amplitude	combinations of three or four measurements in which every telescope's error cancels.
SEFD	a telescope's system-equivalent flux density: lower is more sensitive.
χ^2	misfit between an image's predictions and the measurements, in units of the noise.
Optimism	how much better than the truth a flexible model can fit, by also fitting noise.
Assumption (prior)	the imaging algorithm's built-in preference for some images over others.
Witness / twin pair	two acceptable images on opposite sides of the claim; twins grow from the same starting image.
Separation	how many noise units apart two images' predictions for a future night would be.